TED (21) – REVISION 2021

THIRD SEMESTER DIPLOMA EXAMINATION IN ENGINEERING AND TECHNOLOGY

EMBEDDED SYSTEMS

MODEL QUESTION PAPER

Time: 3 hours Maximum Marks: 75

PART A

I. Answer all the following questions in one word or sentence.

(9 x 1 = 9 Marks)

		Module Outcome	Cognitive level
1	List any two examples of embedded systems	M 1.02	R
2	Give two examples for on-board communication interfaces in a common embedded system hardware	M 1.06	R
3	List any two AVR family microcontrollers	M 2.01	R
4	Name any two timers available in Atmega32	M 2.04	R
5	Name the CPU register which has the address of the next instruction to be executed from memory	M 2.03	U
6	The expansion form of RTC is …………………………	M3.06	R
7	………….. is any motor-driven system with a feedback element built in	M 3.05	U
8	Write the expansion of RTOS	M 4.01	R
9	The term used to represent processes in RTOS for embedded system are called	M 4.02	U

PART B

II. Answer any Eight questions from the following

(8 x 3= 24 Marks)

		Module Outcome	Cognitive level
1	Describe about memory types used in embedded systems	M 1.05	U
2	List any three differences between embedded system and general purpose computer	M 1.01	U
3	Write an AVR embedded C program to read the status of a switch and display the status into an LED	M 2.05	A
4	List the criteria to a microcontroller for an embedded system	M 2.01	U
5	List logic operations in AVR microcontroller	M 2.05	R

6	Draw the diagram for interfacing Relay with AVR	M 3.02	U
7	Explain SPI interfacing	M 3.08	U
8	Illustrate Matrix keyboard interfacing	M 3.04	A
9	Outline kernel in an operating system	M 4.01	R
10	List any six popular RTOS	M4.02	R

PART C

III. Answer all questions from the following (6x 7 = 42 Marks)

		Module Outcome	Cognitive level
1	Classify embedded systems based on any two criteria	M 1.02	U
	OR		
2	Explain architecture of embedded system with simple block diagram	M 1.04	U
3	Describe about Atmega32 with the help of a simplified block diagram	M 2.02	U
	OR		
4	Develop AVR embedded C program to blink an LED connected to a port pin with a delay of 2 milli seconds	M 2.06	A
5	Develop embedded C program for interfacing LCD with AVR	M 3.04	A
	OR		
6	Explain RTC interfacing with AVR	M 3.06	U
7	Develop embedded C program for interfacing of DC motor with AVR	M 3.05	A
	OR		
8	Explain I2C interfacing with AVR	M 3.07	U
9	Explain operating system architecture with simplified block diagram	M 4.01	U
	OR		
10	Explain any three functions of RTOS Kernel	M 4.02	U
11	List the selection criteria for embedded OS	M 4.03	U
	OR		
12	Outline the Micro C/OS-II with its services	M 4.04	U

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Time: 3 hours Maximum Marks: 75

PART A

I. Answer all the following questions in one word or sentence.

(9 x 1 = 9 Marks)

		Module Outcome	Cognitive level
1	List any two applications of embedded systems	M 1.02	R
2	----- is an example for sensor used in an embedded system	M 1.05	R
3	List any two features of Atmega32 microcontroller	M 2.03	R
4	Name any two timers registers in Atmega32	M 2.04	R
5	Define interrupt priority	M 2.07	U
6	I2C is a ……………. communication protocol	M3.07	U
7	------ is a digitally controlled DC motor, that can make movement is steps	M 3.05	U
8	List any two types of operating systems	M 4.01	R
9	…………. in an operating systems allows a user to perform more than one task at atime.	M 4.02	U

PART B

II. Answer any Eight questions from the following

(8 x 3= 24 Marks)

		Module Outcome	Cognitive level
1	Distinguish hardware and software components in an embedded system	M 1.03	U
2	Describe the purpose of embedded systems	M 1.02	U
3	Write an AVR embedded C program to read the status of a switch and activate a relay accordingly	M 2.05	A
4	Write a time delay function in embedded C and explain delay calculation	M 2.06	A
5	Explain hardware interrupt features in AVR	M 2.07	U

6	Illustrate opto coupler interfacing with AVR with diagram	M 3.02	A
7	OutlineADC interfacing with AVR	M 3.06	U
8	Illustrate Servo motor interfacing with AVR	M 3.05	A
9	Explain task scheduling in embedded systems	M 4.02	U
10	List selection criteria for RTOS	M4.03	U

PART C

III. Answer all questions from the following (6x 7 = 42 Marks)

		Module Outcome	Cognitive level
1	List differences between general purpose computer and embedded system	M 1.01	U
	OR		
2	Describe different types of communication interfaces in an embedded system	M 1.06	U
3	Describe features of AVR family of microcontrollers	M 2.01	U
	OR		
4	Develop AVR embedded C program to display the status of a port pin into an LED connected to another pin	M 2.05	A
5	Explain registers associated with AVR ports	M 2.03	U
	OR		
6	Write embedded C program to toggle a port bit with a time delay, using timer 0 operation	M 2.06	A
7	Develop AVR embedded C program to display 00 and 99 with a 1sec delay in a seven segment display connected to its port	M 3.03	A
	OR		
8	Explain Matrix Keyboard interfacing with AVR	M 3.04	U
9	Develop embedded C program for interfacing of stepper motor with AVR	M 3.05	A
	OR		
10	Explain SPI interfacing with AVR	M 3.08	U
11	Explain task communication and task synchronization in embedded system	M 4.02	U
	OR		
12	Describe the features of Micro C/OS-II	M 4.04	U